

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 05

Miscellaneous Topics



By- Chandan Gupta Sir

Recap of Previous Lecture



Number System

PI, EPI





Topics to be Covered

State transition diagram





#Q.

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	1	1	
$\bar{A}B$	1	X		1
AB	1	X		1
$A\bar{B}$		1	1	

$PI_s \rightarrow B\bar{D}, \bar{B}D, \bar{A}\bar{C}, \bar{C}D, B\bar{C}$

$EPI_s \rightarrow B\bar{D}, \bar{B}D, \bar{A}\bar{C}$

Non-EPI : $\bar{C}D, B\bar{C}$



#Q.

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1		X	1
$\bar{A}B$	1	1	1	1
AB				X
$A\bar{B}$	1			1

H.W.

$PI_s \rightarrow$

$EPI_s \rightarrow$



#Q.

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1		1	1
$\bar{A}B$	X			
AB	1		1	X
$A\bar{B}$	1		1	1

H.W.

$PI_s \rightarrow$

$EPI_s \rightarrow$



#Q.

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	1		
$\bar{A}B$	1	1		
AB	X	1	1	1
$A\bar{B}$	X			

H.W.

$PI_s \rightarrow$

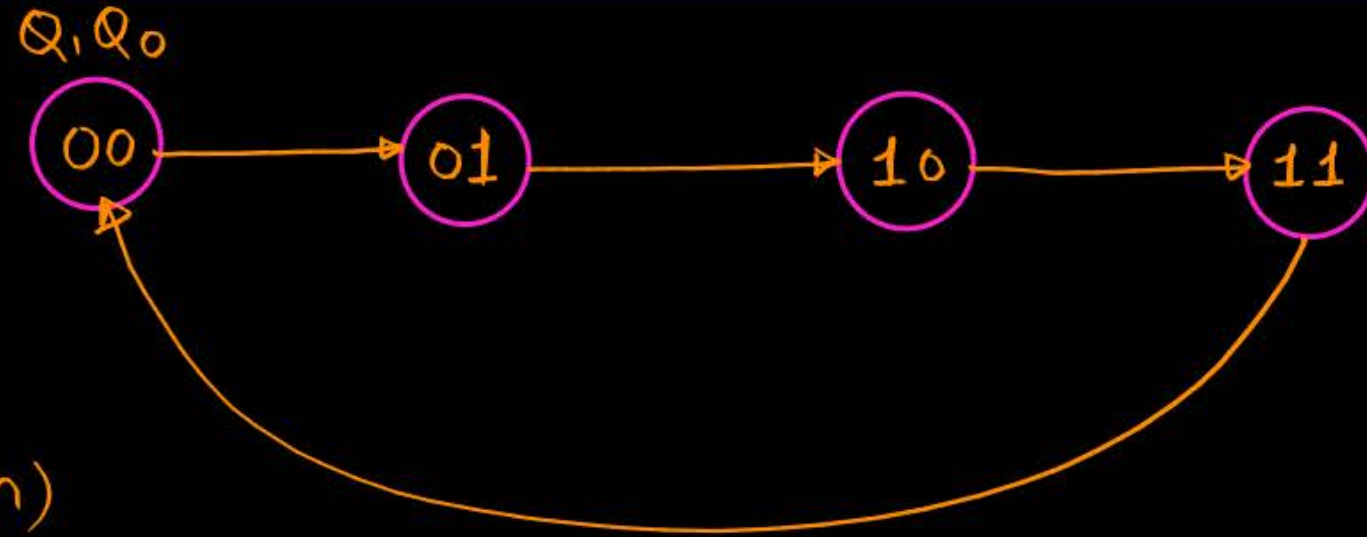
$EPI_s \rightarrow$

[State Transition Diagram]



→ What is state transition diagram ?

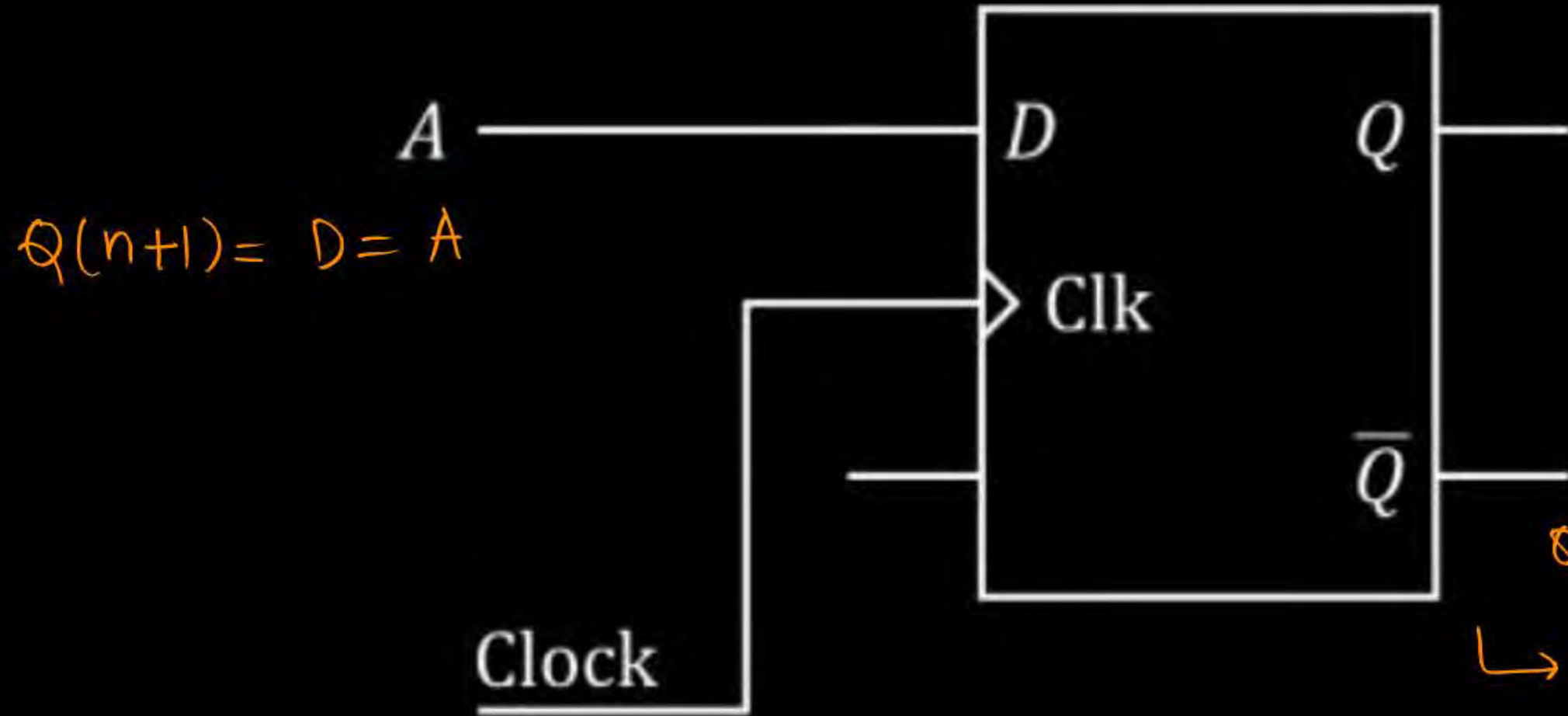
- The diagram that represents all the possible transitions b/w states of a sequential CKT.



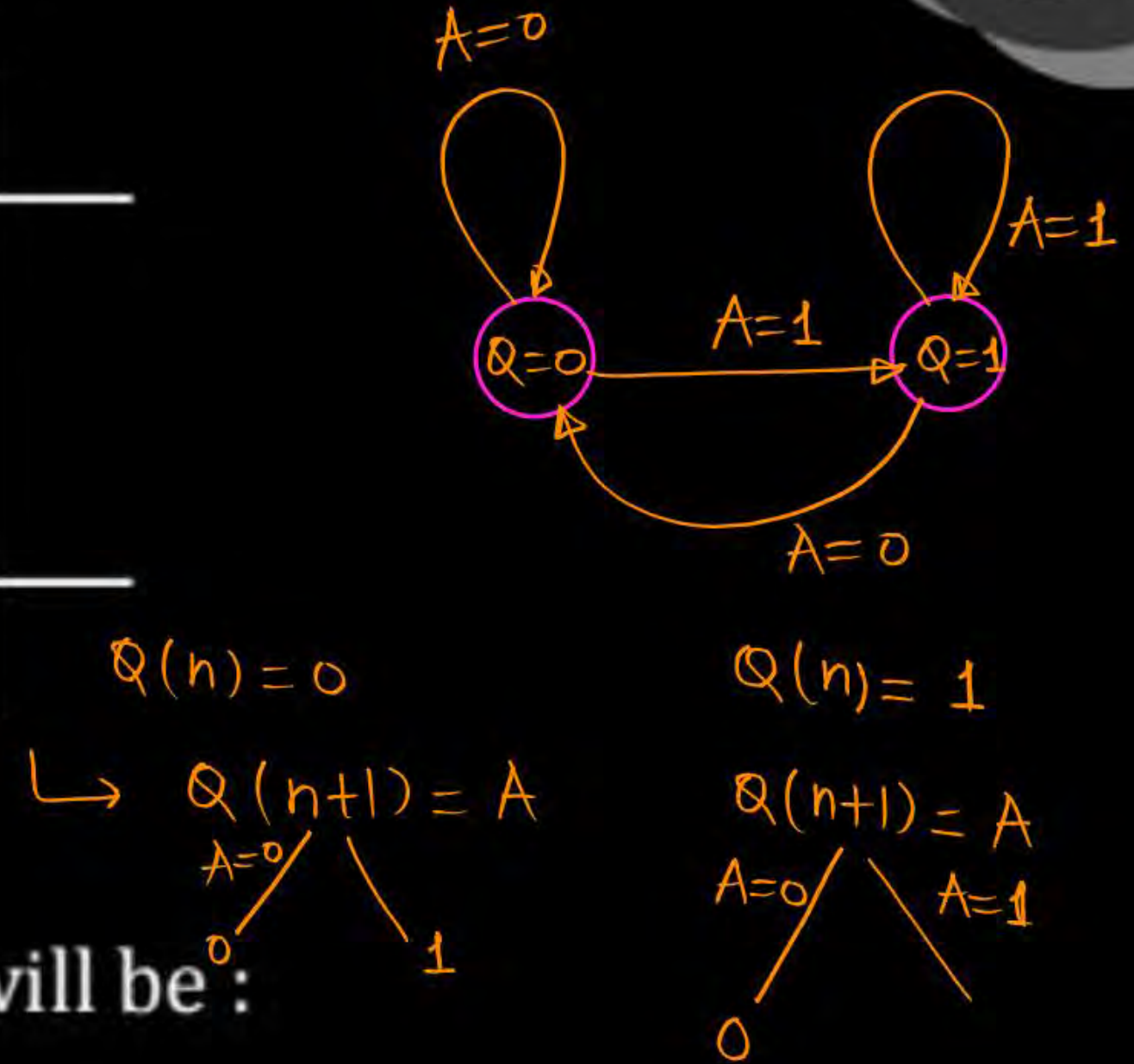
$$Q_0(n+1) = \overline{Q_0(n)}$$

$$Q_1(n+1) = Q_0(n) \oplus Q_1(n)$$

- Lets understand it with an example :



State transition diagram for above circuit will be :



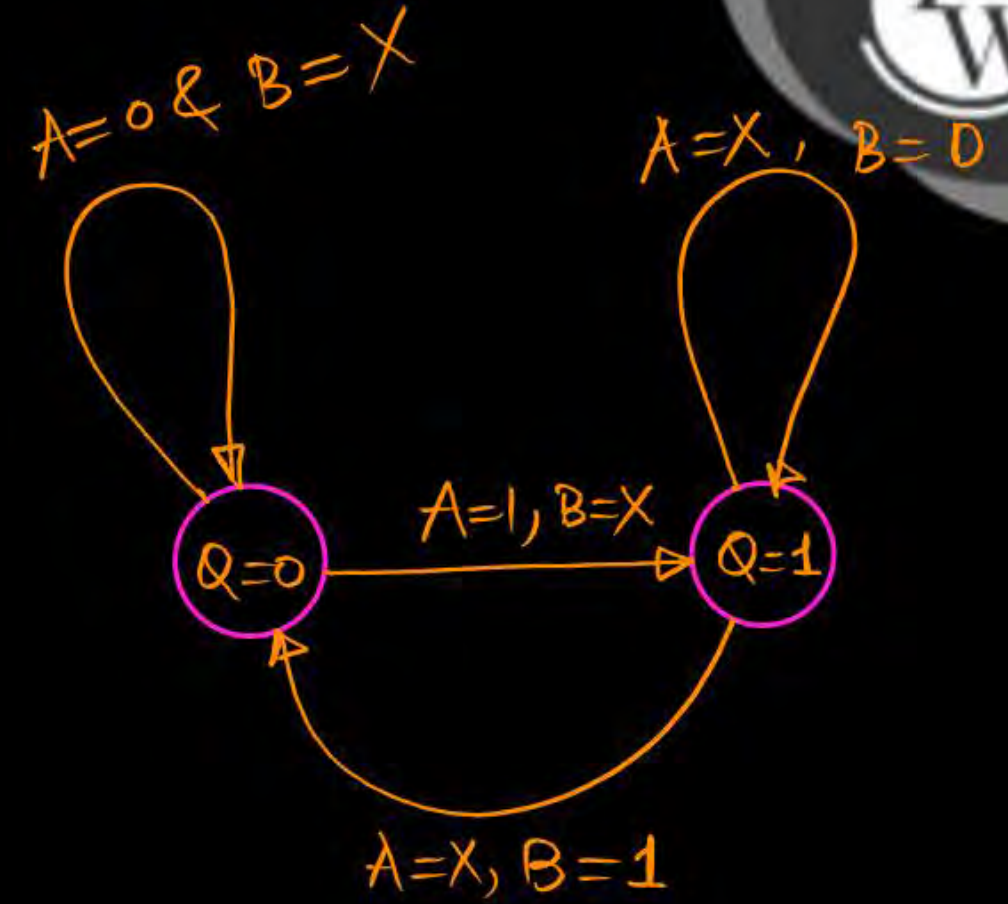
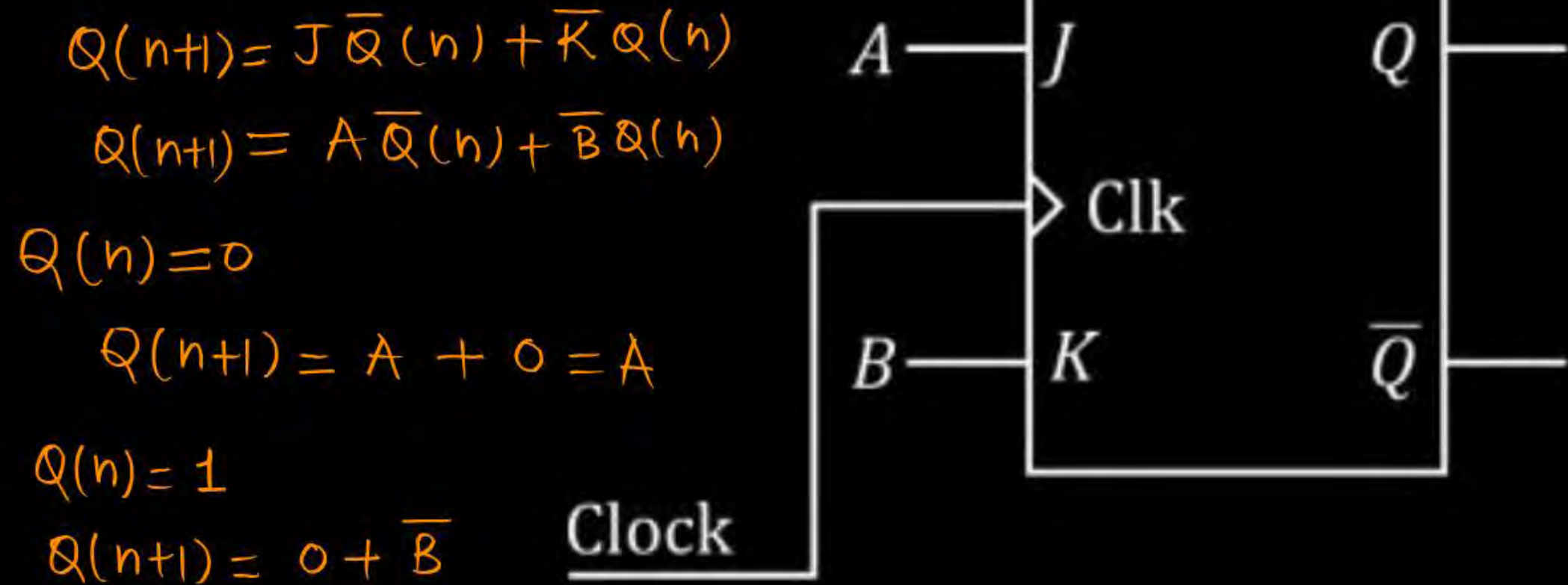
A	Q	$Q(n+1)$
0	0	0
0	1	0
1	0	1
1	1	1

$$Q(n+1)[A, Q] = \Sigma(2, 3)$$

	$\bar{Q}(n)$	$Q(n)$	
$\bar{A}$			
A	1	1	→ A

$Q(n+1) = A$

- A sequential circuit is as given below :

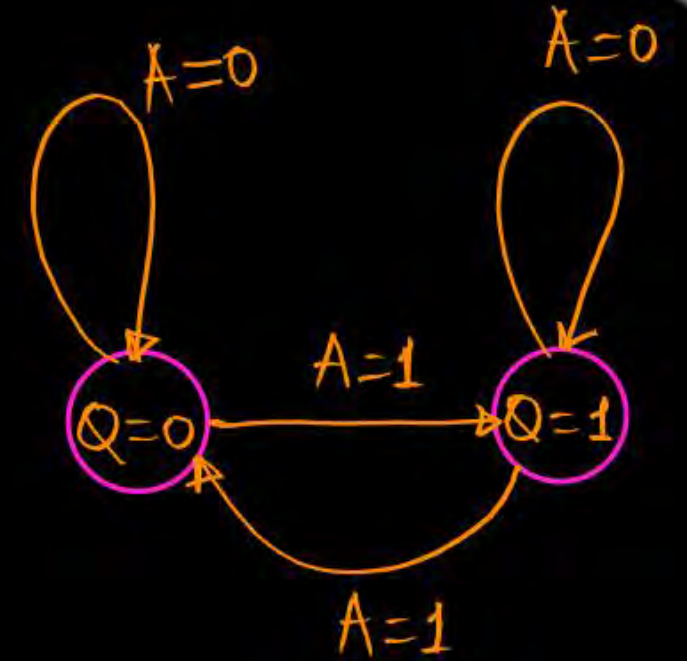
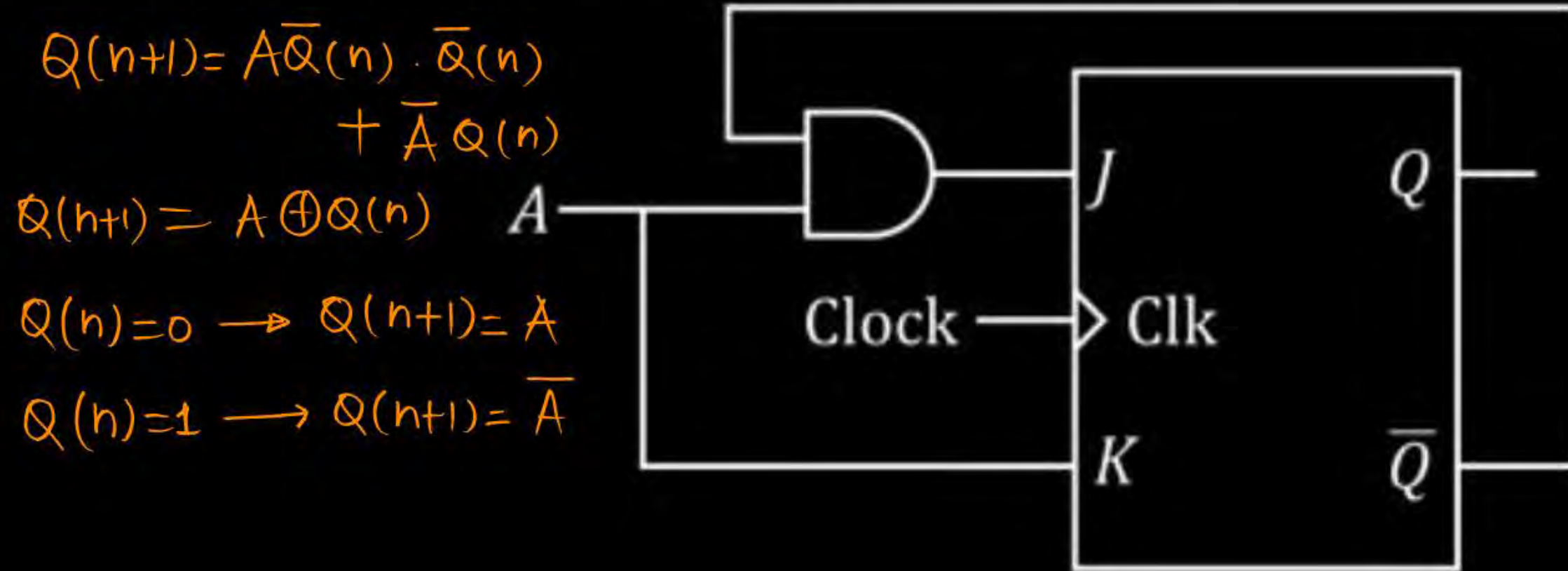


H.W.

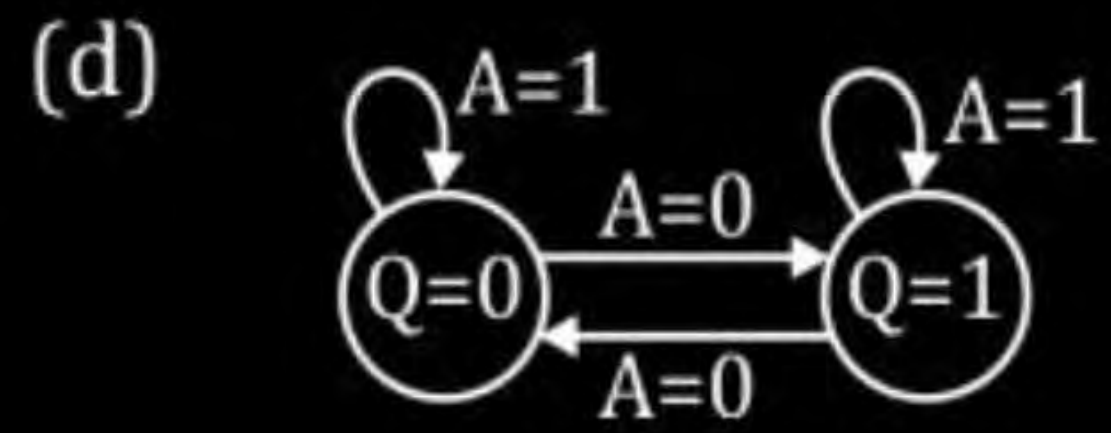
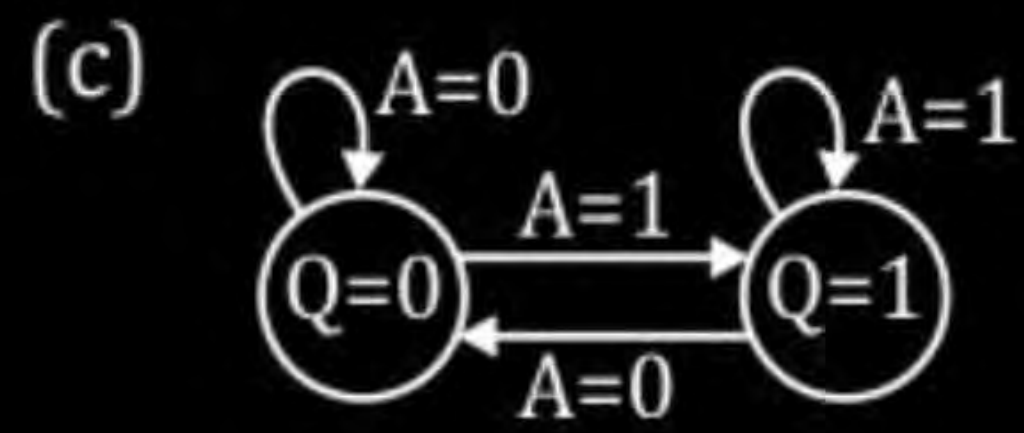
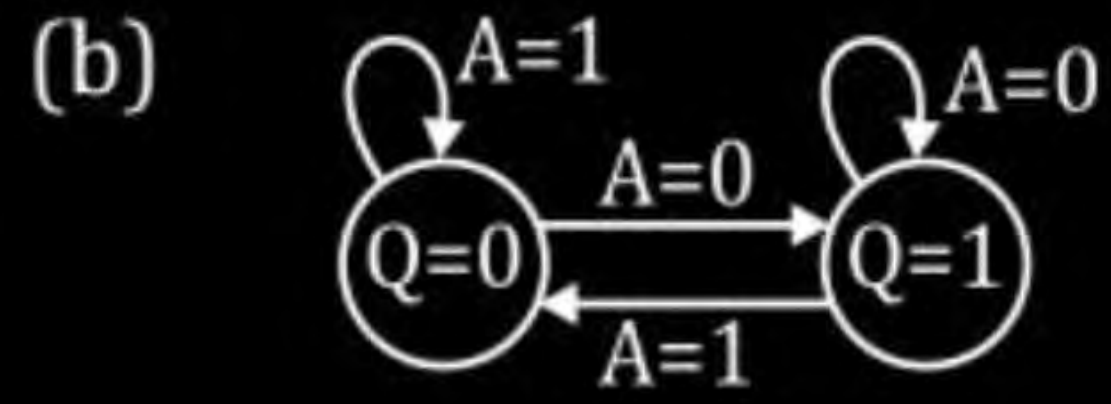
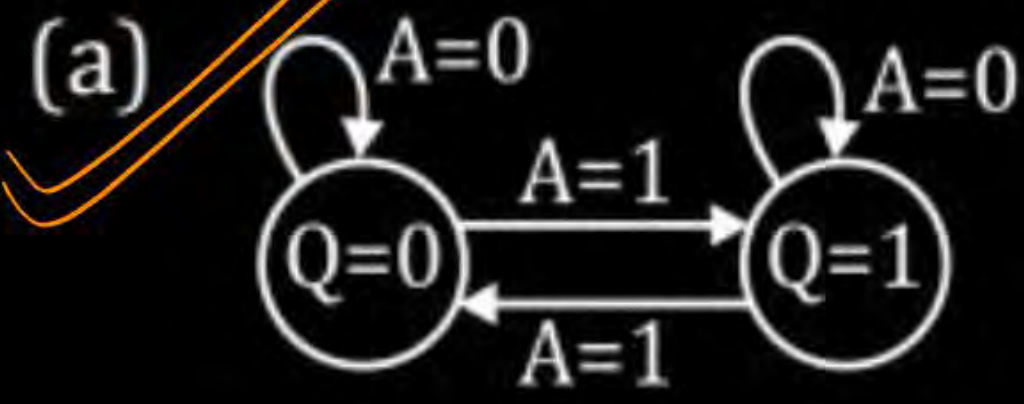
State transition diagram of above circuit will be :

[Question]

A sequential circuit is as given below :

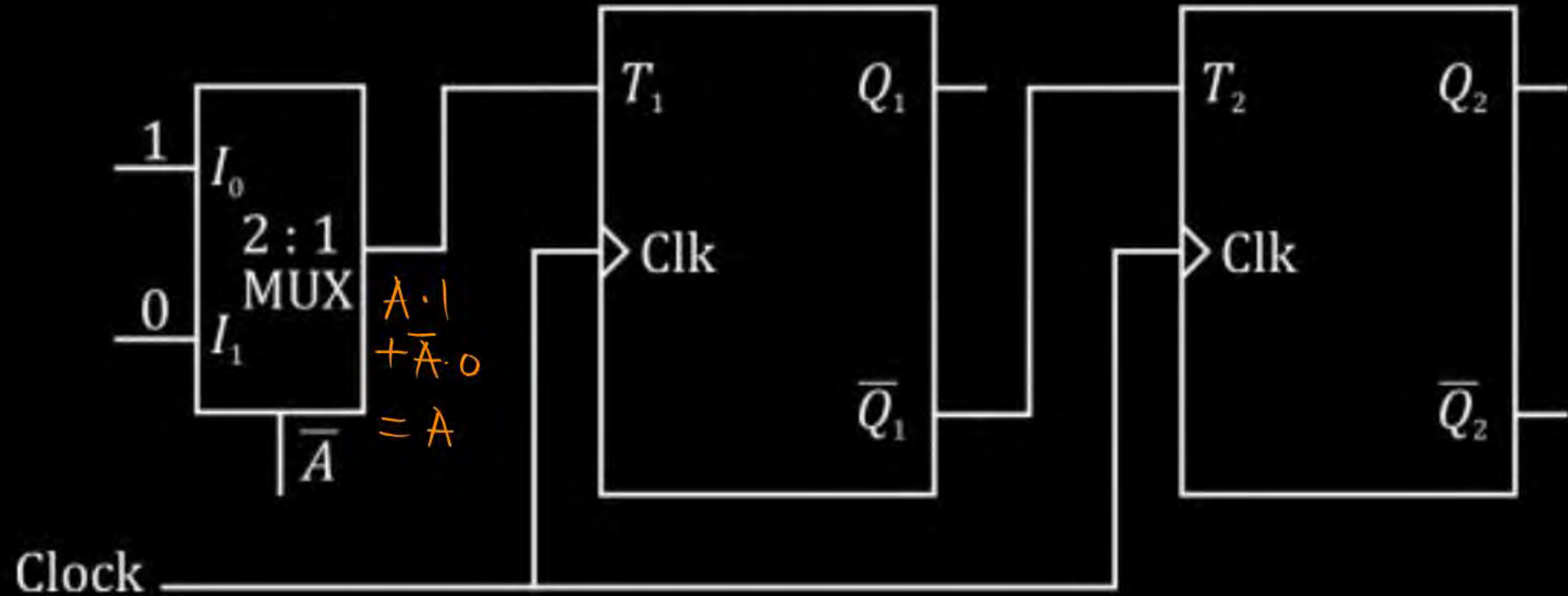


State transition diagram of above sequential circuit is

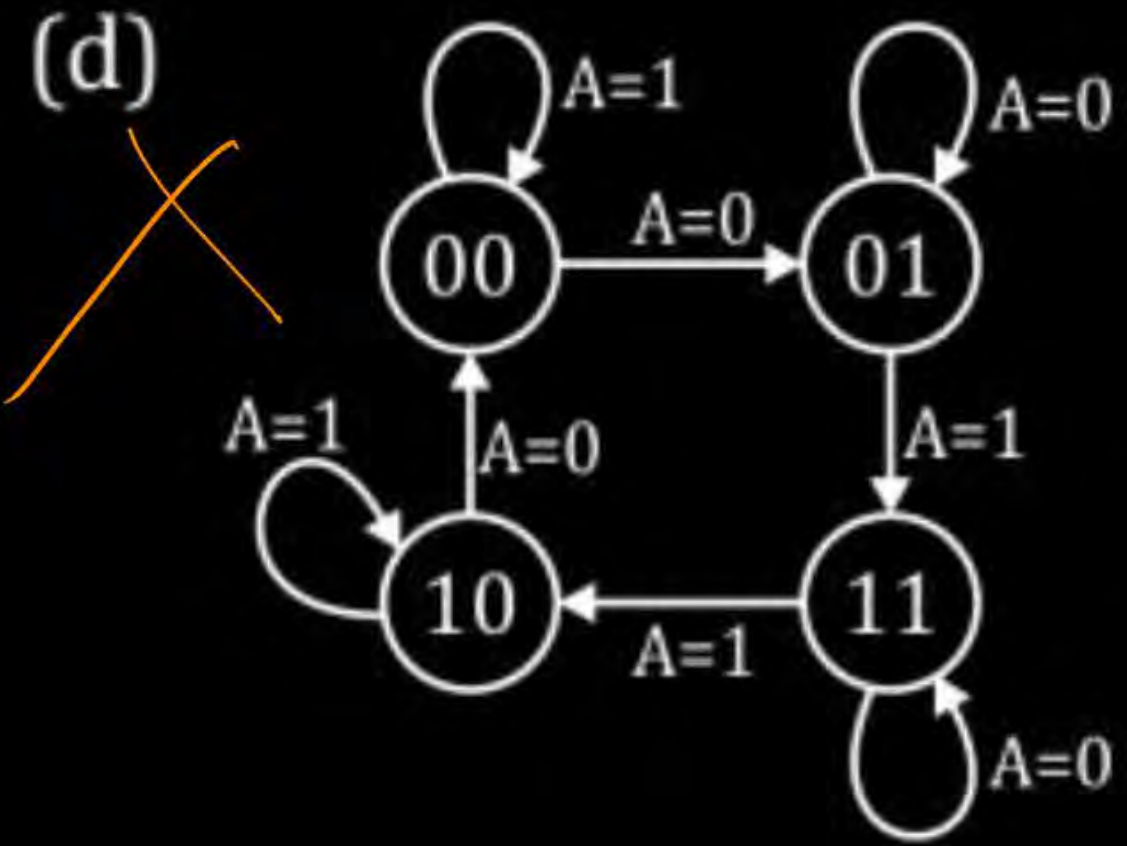
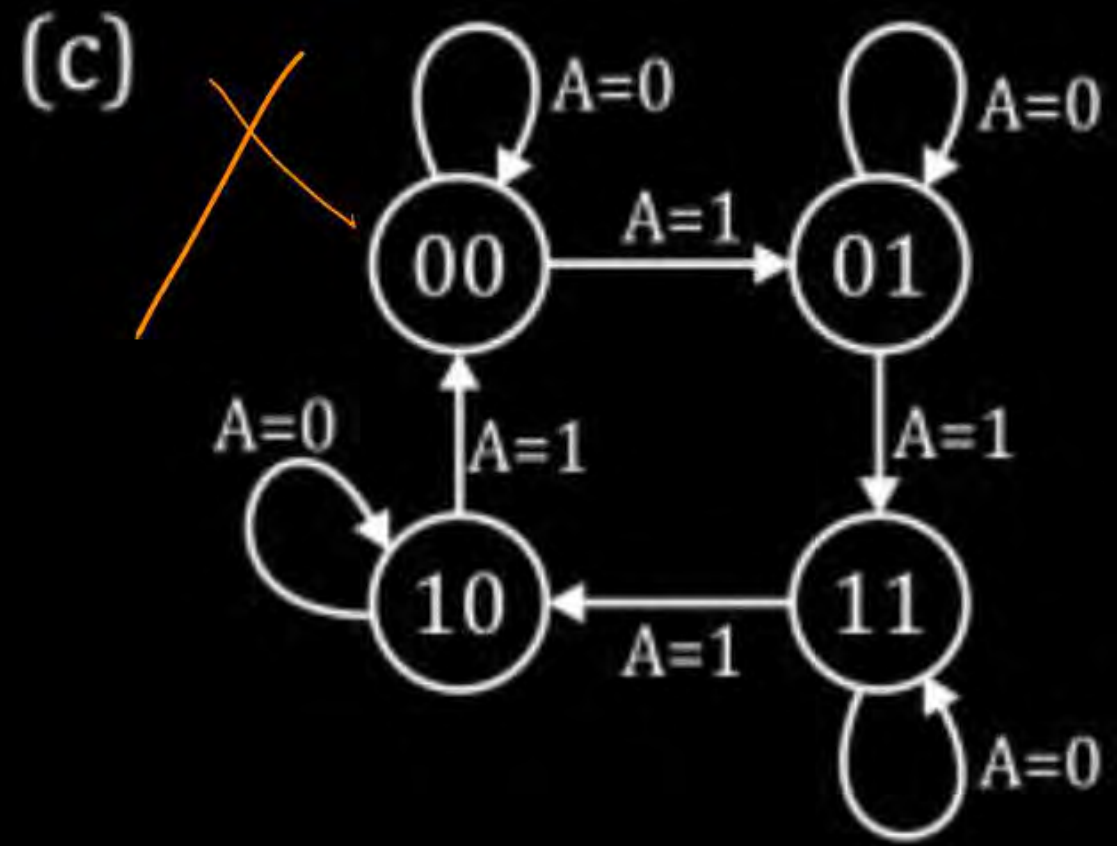
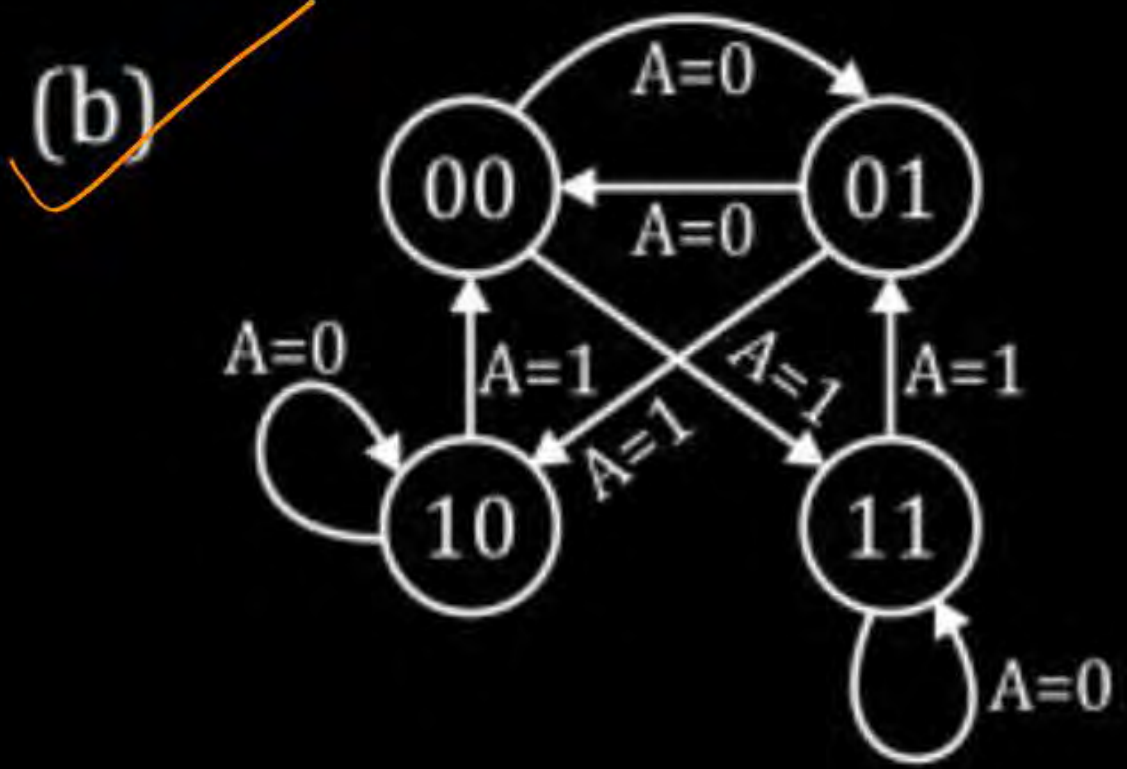
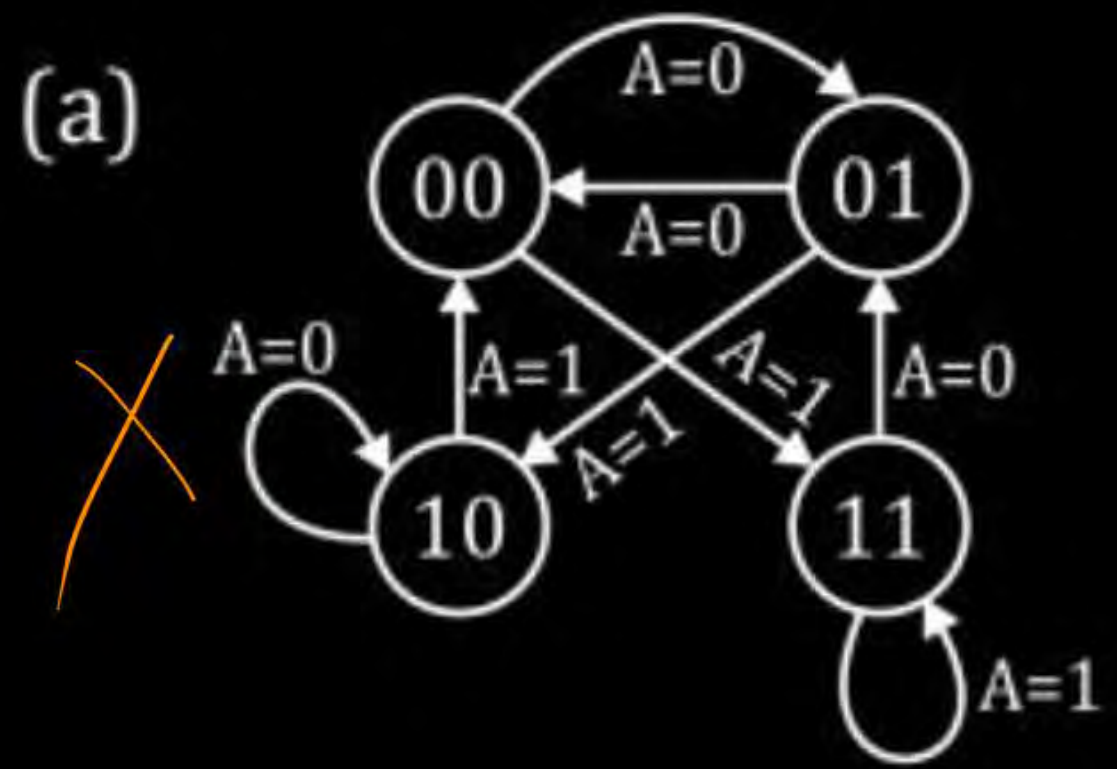


[Question]

A sequential circuit is as given below :

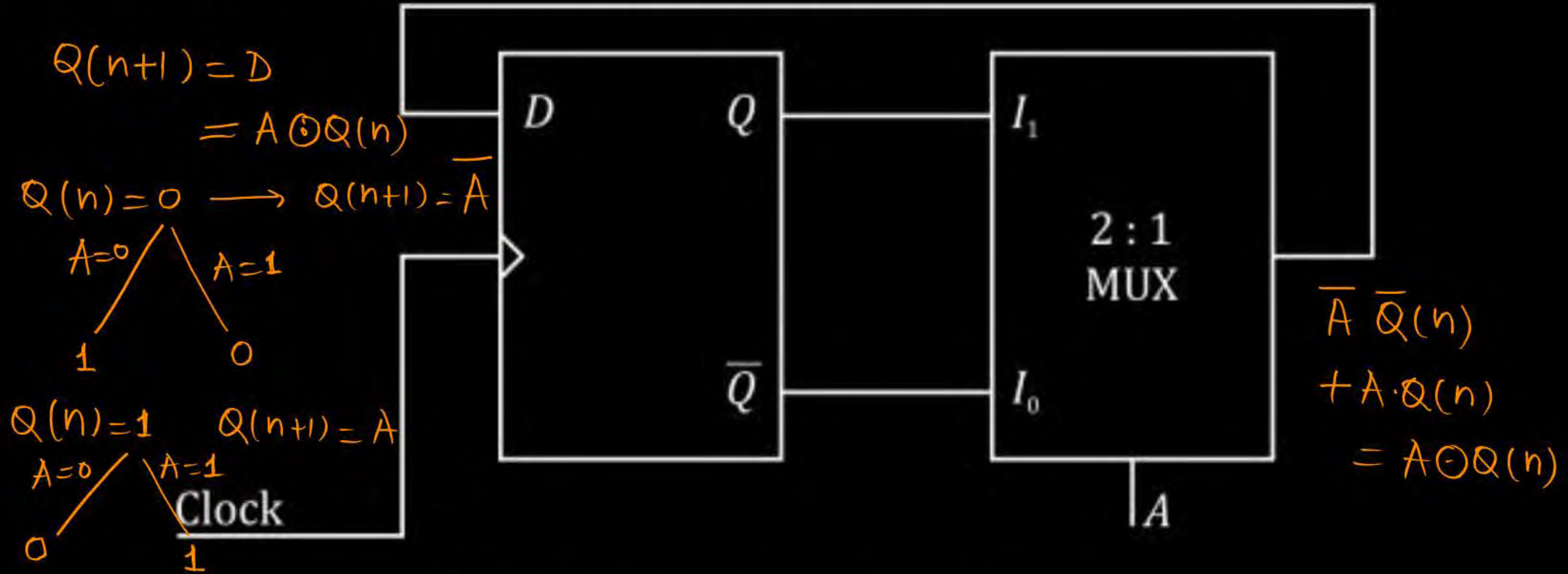


State transition diagram of above circuit will be :

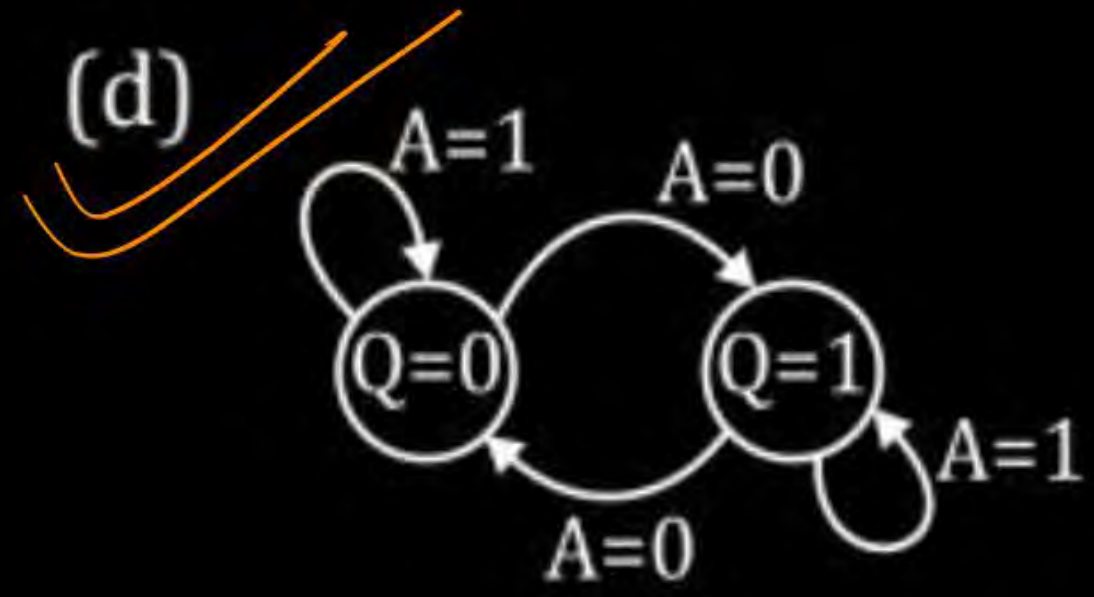
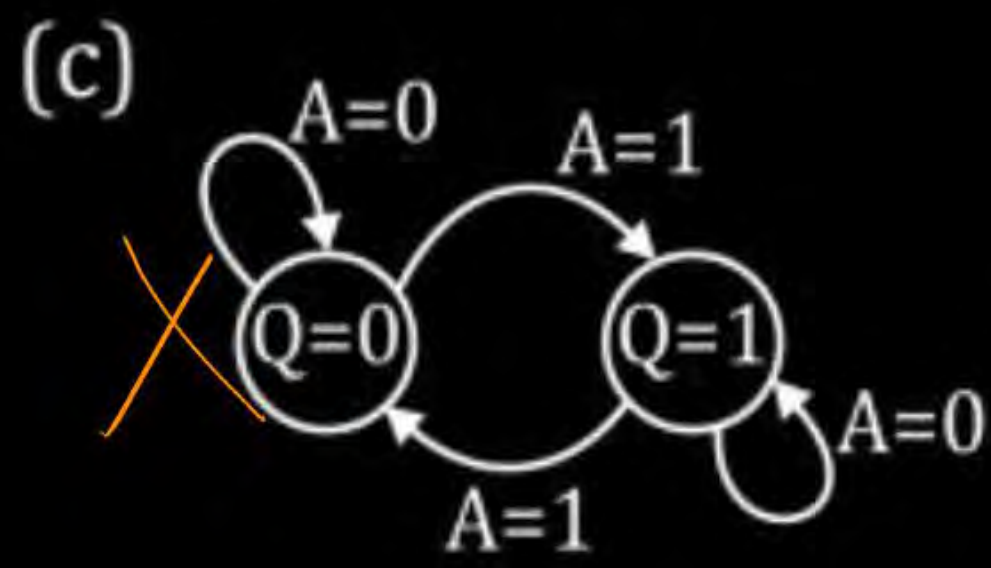
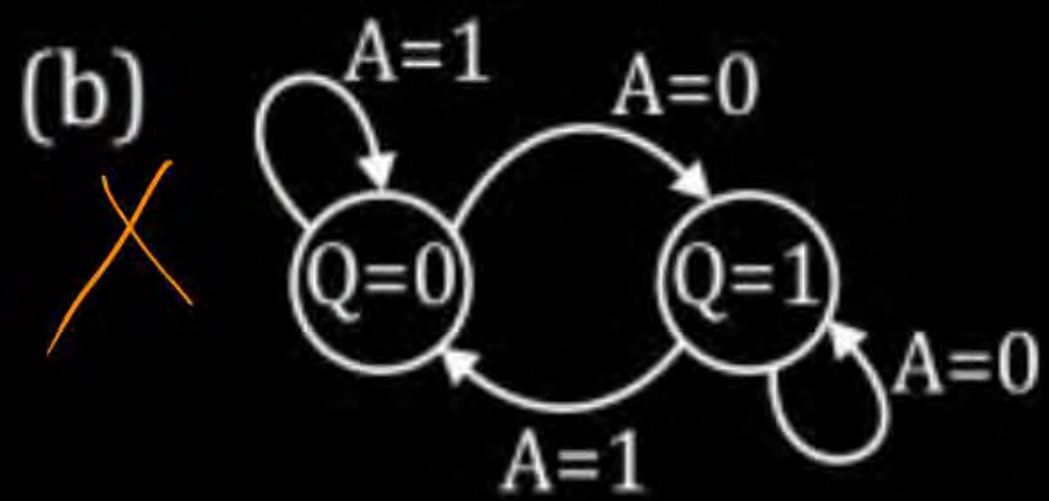
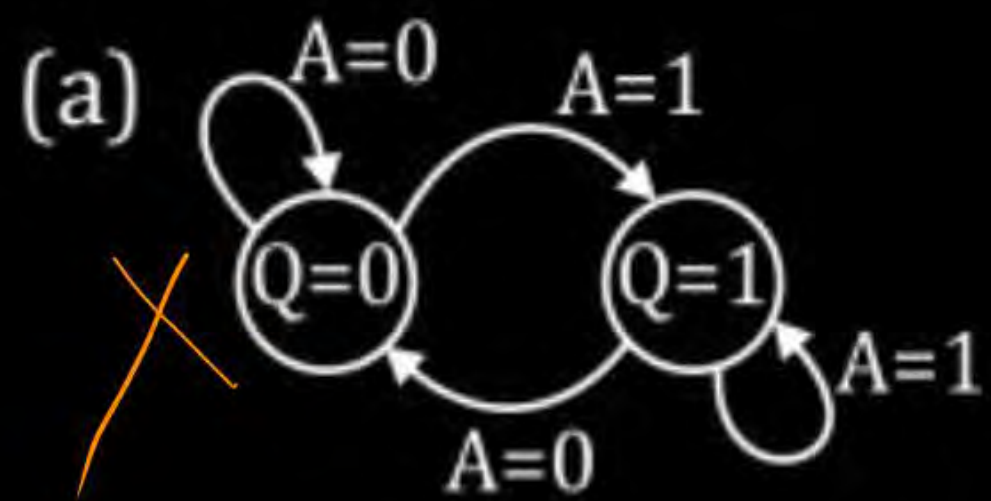


[Question]

A sequential circuit is as given below :

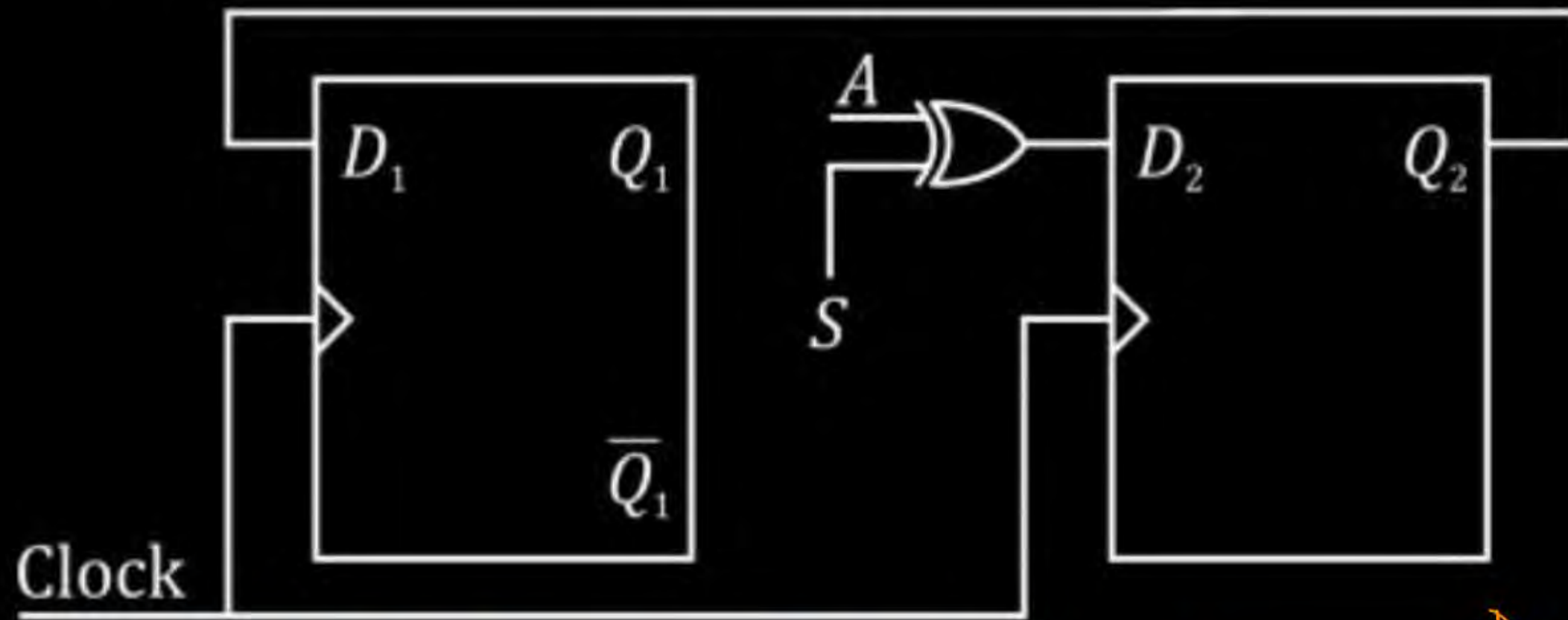


State transition diagram of above circuit will be :



[Question]

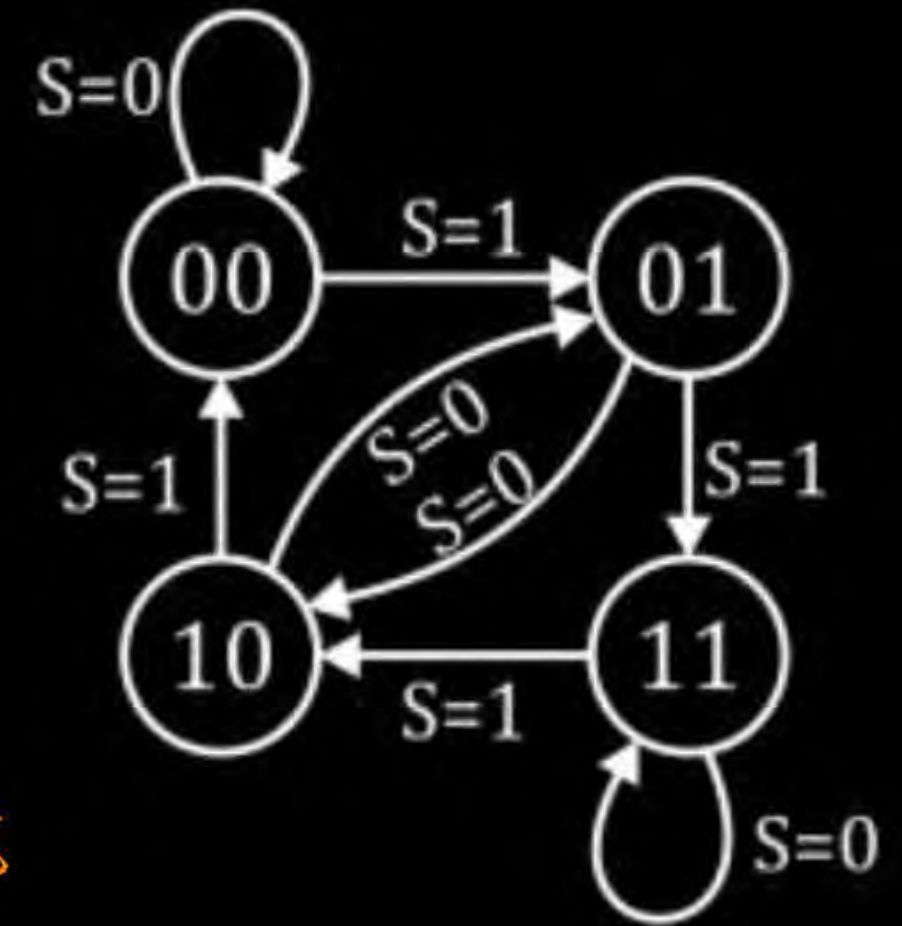
The digital logic shown in the figure satisfies the given state diagram when Q_1 is connected to input A of the XOR gate :



$$D_2 = A \oplus S$$

$$Q_1(n+1) = D_1 = Q_2(n) \quad Q_2(n+1) = D_2 = A \oplus S$$

$$A = Q_1(n) \rightarrow Q_2(n+1) = Q_1(n) \oplus S$$



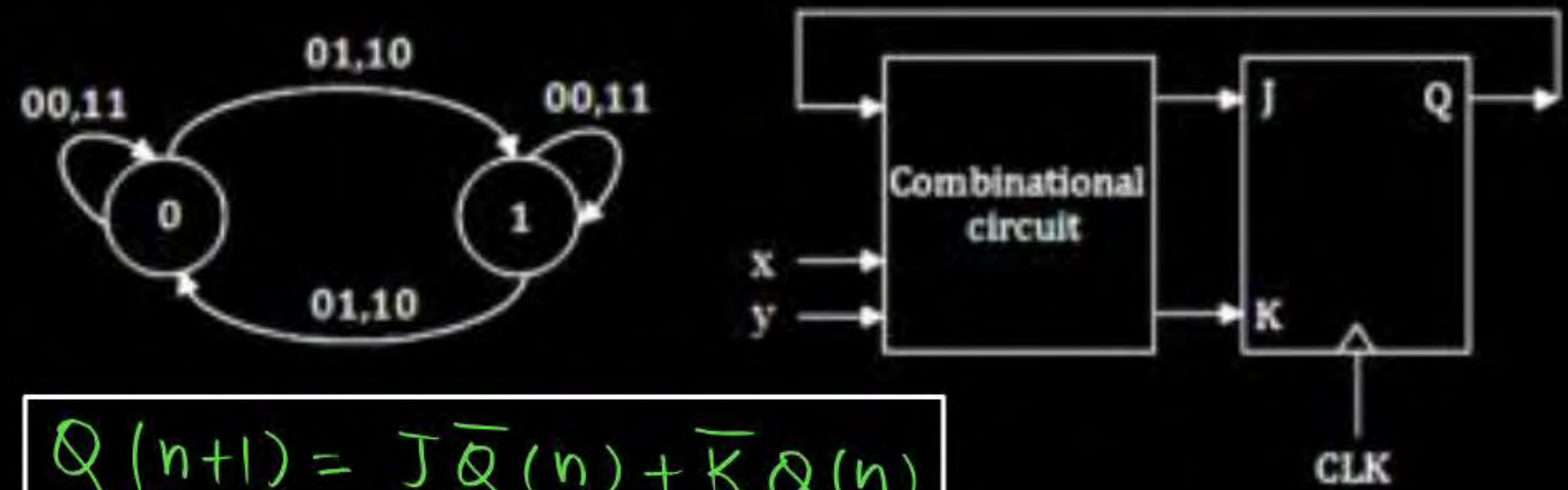
Suppose XOR gate is replaced by XNOR gate then which of the option preserves the state diagram :

- (a) Input A is connected to $\bar{Q}_2$ ✗
- (b) Input A is connected to Q_2 ✗
- (c) Input A is connected to $\bar{Q}_1$ and S is complimented. ✗
- (d) ✓ Input A is connected to $\bar{Q}_1$ ✓

[MCQ]

Consider the following state diagram and its realization by a JK flip flop :
The combinational circuit generates J and K in terms of x, y and Q. The Boolean expressions for J and K are :

- A ~~$(x \oplus y)'$ and $(x \oplus y)'$~~
- B ~~$(x \oplus y)'$ and $x \oplus y$~~
- C ~~$x \oplus y$ and $(x \oplus y)'$~~
- D $x \oplus y$ and $x \oplus y$



$$Q(n+1) = J\bar{Q}(n) + \bar{K}Q(n)$$

$$\begin{aligned} xy = 00, 11, & \quad Q(n+1) = Q(n) \\ xy = 01, 10 & \quad Q(n+1) = \bar{Q}(n) \end{aligned}$$

x	y	$Q(n)$	$Q(n+1)$
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

$$Q(n+1)[x, y, Q(n)] = \Sigma(1, 2, 4, 7)$$

$$Q(n+1) = \underline{x \oplus y \oplus Q(n)}$$

$$= A \oplus Q(n)$$

$$J = x \oplus y$$

$$K = x \oplus y$$

Parity generator & Parity checker.



Parity generator

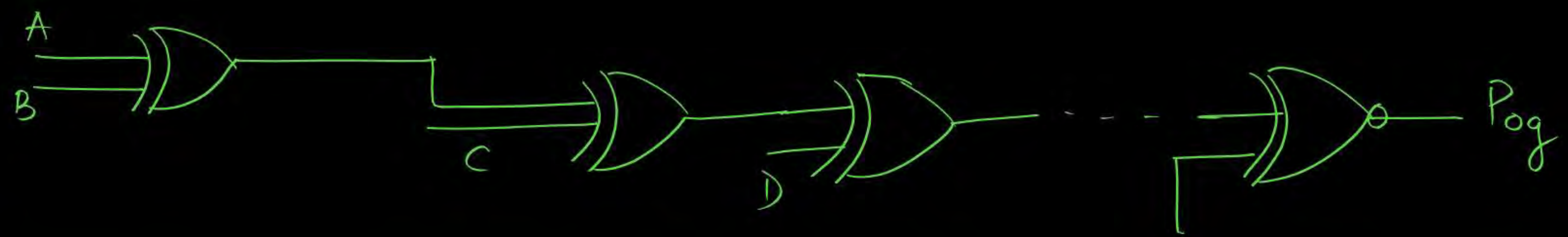
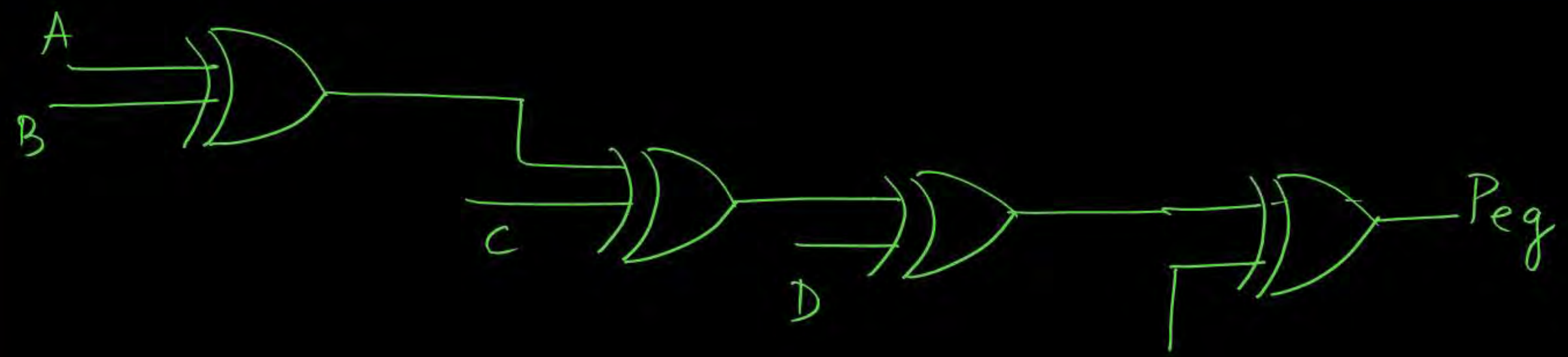
$$P_{eg}[A, B, C, D, \dots]$$

$$= A \oplus B \oplus C \oplus D \dots$$

$$P_{og}[A, B, C, D, \dots]$$

$$= A \oplus B \oplus C \oplus D \dots$$

A	B	C	P_{eg}	P_{og}
0	0	0	0	1
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	0



Parity checker

A B C P P_{ec} P_{oc}

$$P_{ec}[A, B, C, P] = A \oplus B \oplus C \oplus P$$

$$P_{ec}[A, B, C, D, \dots, P] = A \oplus B \oplus C \oplus D \oplus \dots \oplus P$$

$$P_{oc}[A, B, C, P] = \overline{A \oplus B \oplus C \oplus P}$$

$$P_{oc}[A, B, C, D, \dots, P] = \overline{A \oplus B \oplus C \oplus D \oplus \dots \oplus P}$$





Topic : 2 Min Summary

- State transition Diagram
- Parity generator & Parity Checker.

Thank you

GW
Soldiers !

